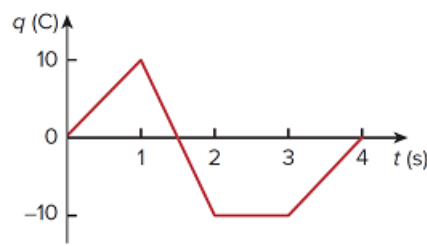


Problem

The charge flowing in a wire is plotted in Fig. 1.24. Sketch the corresponding current.

Figure 1.24:



Step-by-step solution

Step 1 of 4

Refer to Figure 1.24 in the textbook.

Use the waveform in Figure 1.24, to write the equation for charge from $t = 0$ to 1 s.

The response is a straight line between the points $(0, 0)$ to $(1 \text{ s}, 10 \text{ C})$. Therefore, write a straight line equation to get an expression for the charge.

$$q - 0 = \left(\frac{10 - 0}{1 - 0} \right) (t - 0)$$

$$q = 10t$$

Calculate the current for $0 < t < 1$.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(10t) \\ &= 10 \text{ A} \quad \text{for } 0 < t < 1 \end{aligned}$$

Step 2 of 4

Write the equation for charge from $t = 1$ to 2 s.

The response is a straight line between the points $(1 \text{ s}, 10 \text{ C})$ to $(2 \text{ s}, -10 \text{ C})$. Therefore, write a straight line equation to get an expression for the charge.

$$q - 10 = \left(\frac{-10 - 10}{2 - 1} \right) (t - 1)$$

$$q = -20t + 20 + 10$$

$$q = -20t + 30$$

Calculate the current for $1 < t < 2$.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(-20t + 30) \\ &= -20 \text{ A} \end{aligned}$$

Step 3 of 4

For $2 < t < 3$, the charge equation is, $q = 3$.

Calculate the current for $2 < t < 3$.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(3) \\ &= 0 \text{ A} \end{aligned}$$

Write the equation for charge from $t = 3$ to 4 s.

The response is a straight line between the points $(3 \text{ s}, -10 \text{ C})$ to $(4 \text{ s}, 0 \text{ C})$. Therefore, write a straight line equation to get an expression for the charge.

$$\begin{aligned} q + 10 &= \left(\frac{0 + 10}{4 - 3} \right) (t - 3) \\ q &= 10t - 30 - 10 \\ q &= 10t - 40 \end{aligned}$$

Calculate the current for $3 < t < 4$.

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(10t - 40) \\ &= 10 \text{ A} \end{aligned}$$

Step 4 of 4

Therefore, the value of the current at different intervals is,

$$i = \begin{cases} 10 \text{ A} & \text{for } 0 < t < 1 \\ -20 \text{ A} & \text{for } 1 < t < 2 \\ 0 \text{ A} & \text{for } 2 < t < 3 \\ 10 \text{ A} & \text{for } 3 < t < 4 \end{cases}$$

Draw the plot for current.

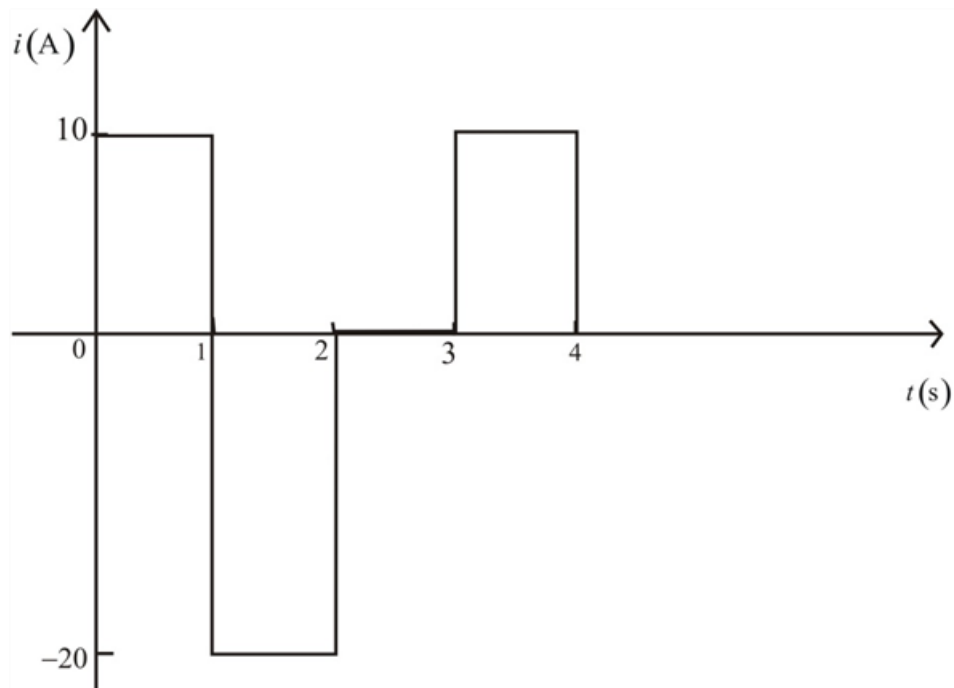


Figure 1